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(54) **BATTERY MANAGEMENT SYSTEM HAVING HIERARCHICAL STRUCTURE AND METHOD FOR OPERATING SAME**

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ABSTRACT

A battery management system (BMS) having a hierarchical structure may include a plurality of slave BMSs and a master BMS configured to interwork with the plurality of slave BMSs and to monitor a battery group including a plurality of batteries. The master BMS may be configured to collect state information of one or more batteries from the plurality of slave BMSs and to collect state information of the battery group and the group state information is collected through a serial communication network.

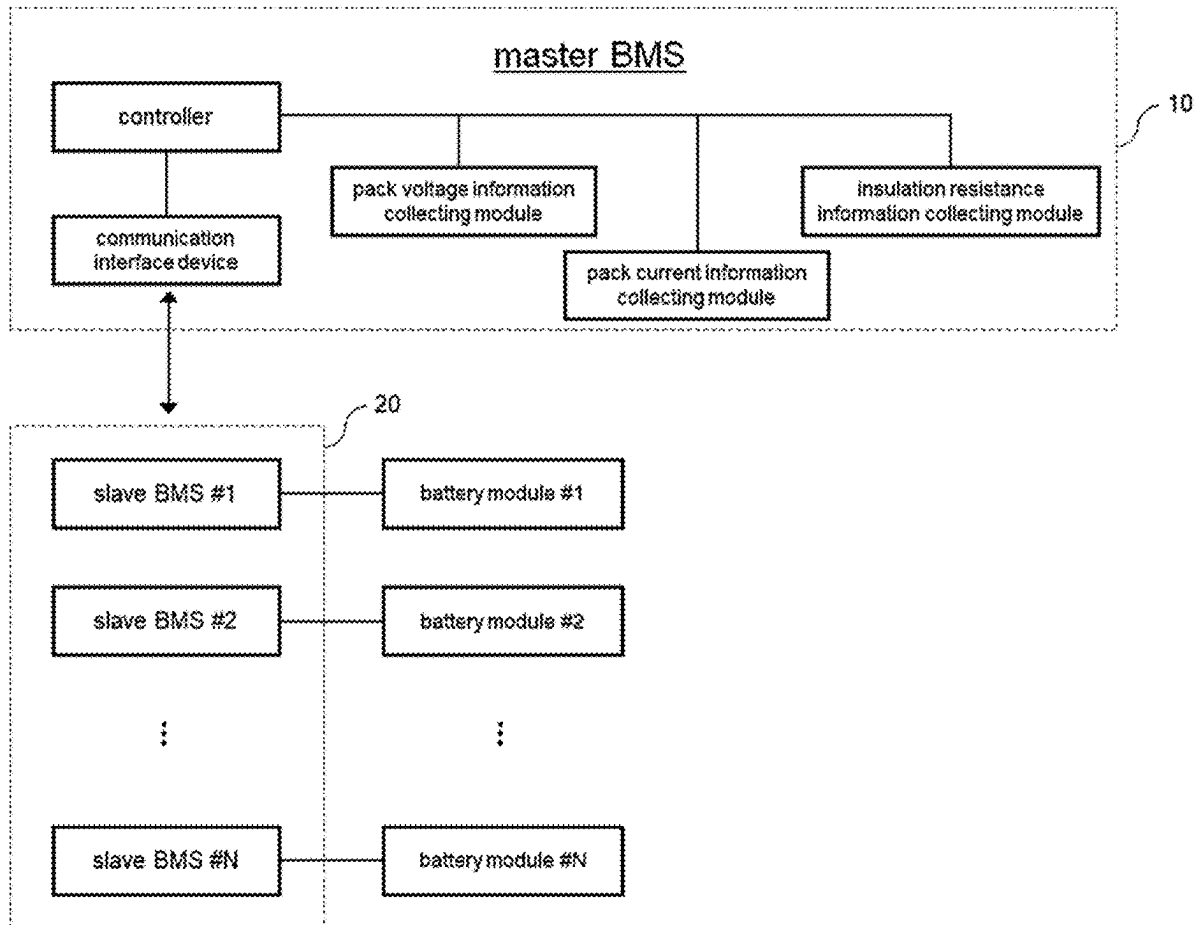
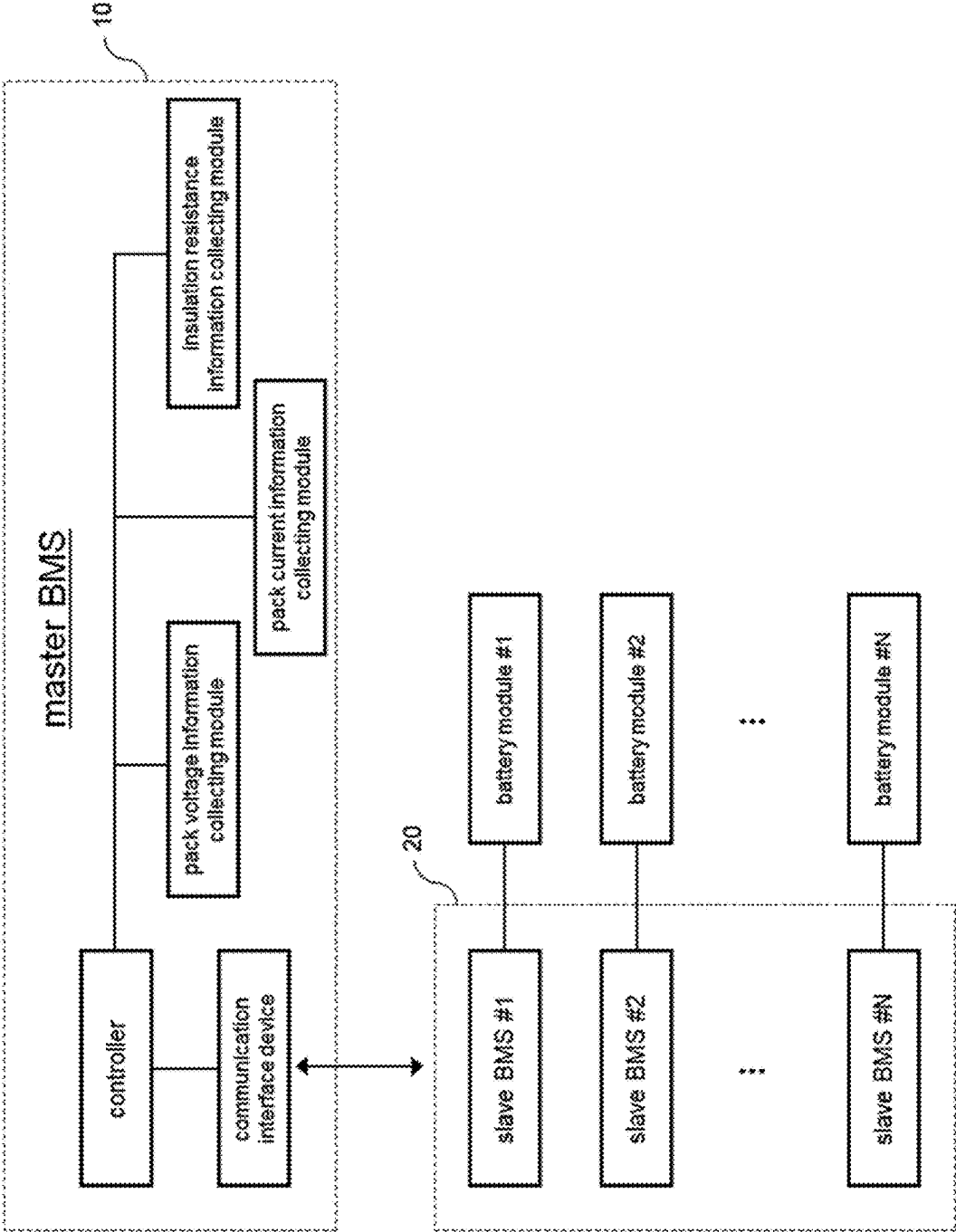
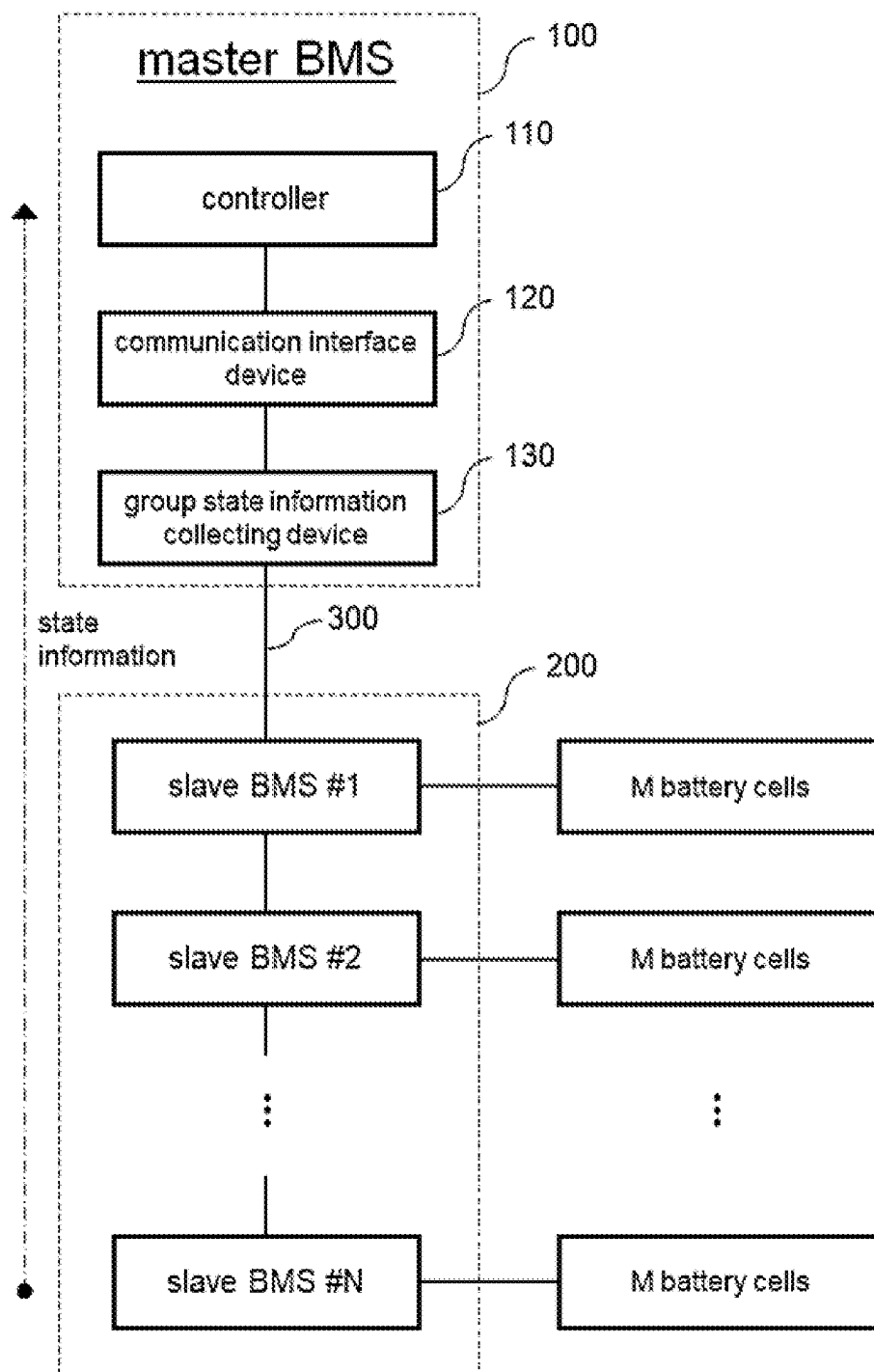


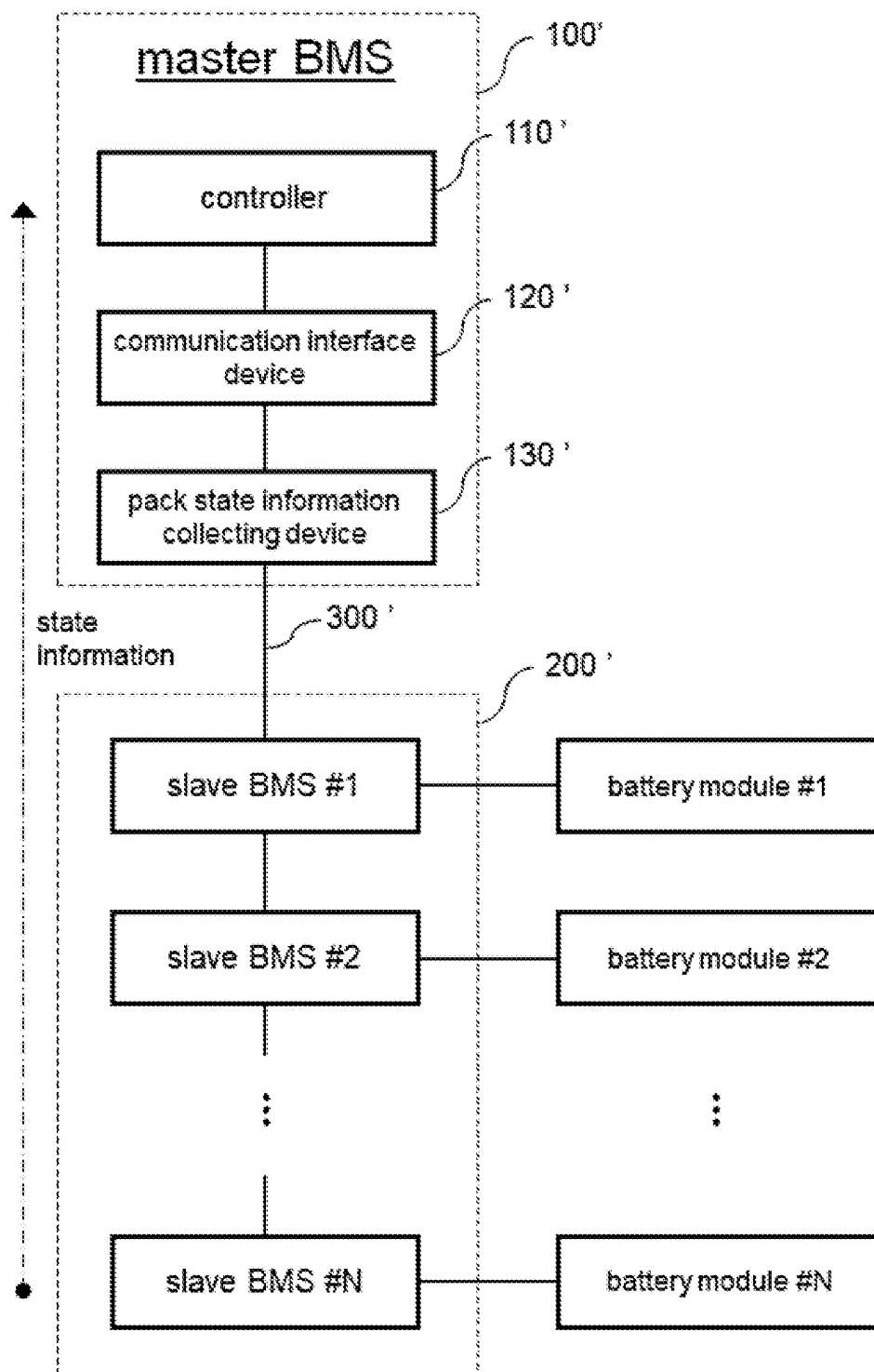
Figure 1



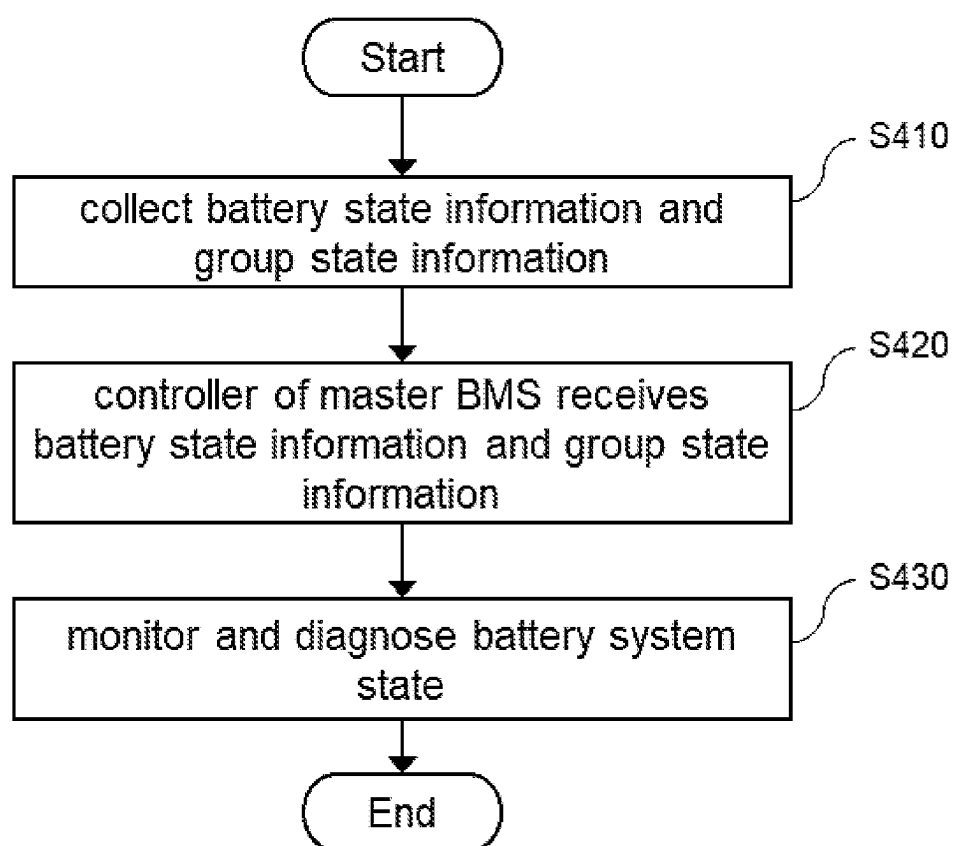
[Figure 2]



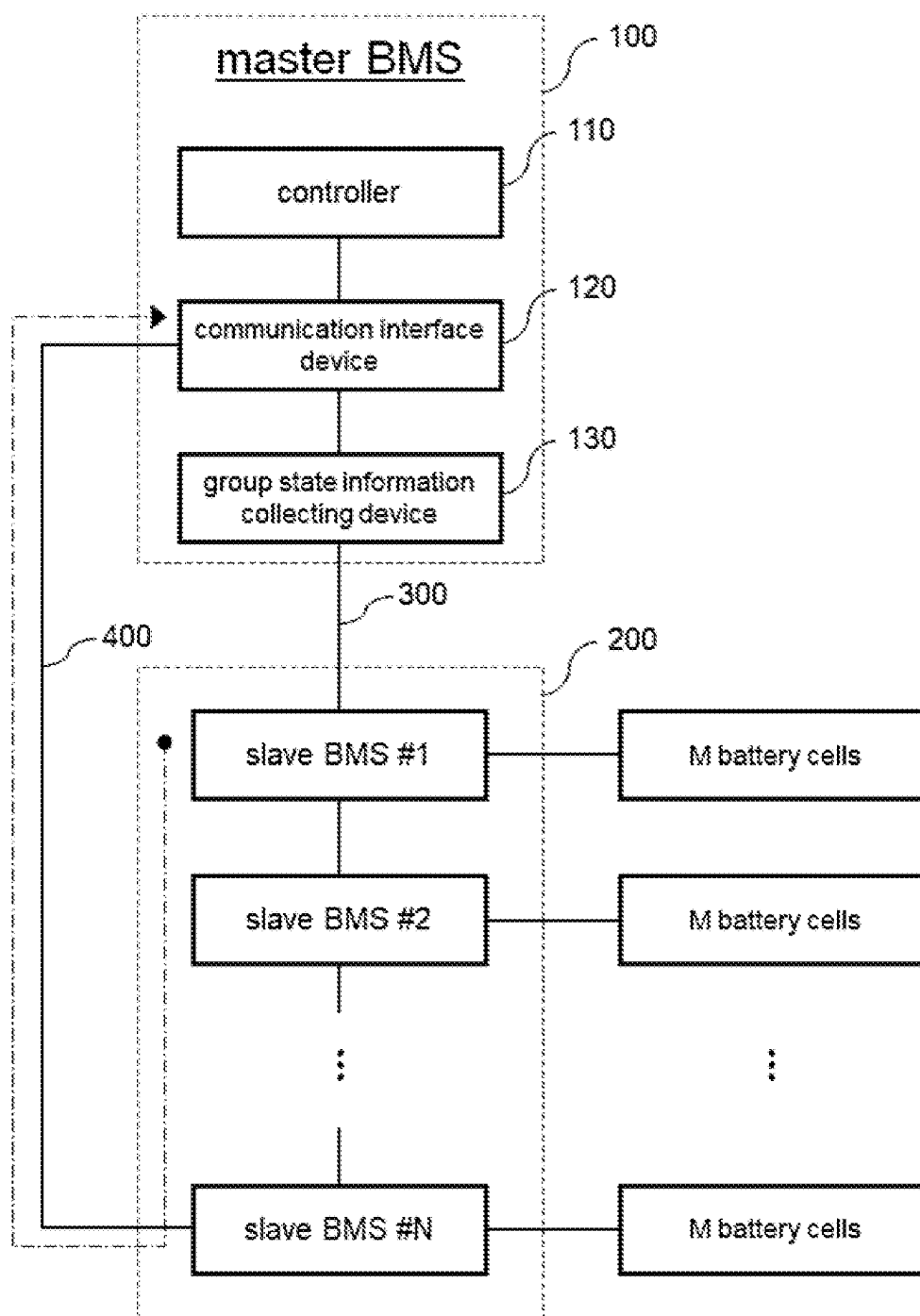
[Figure 3]



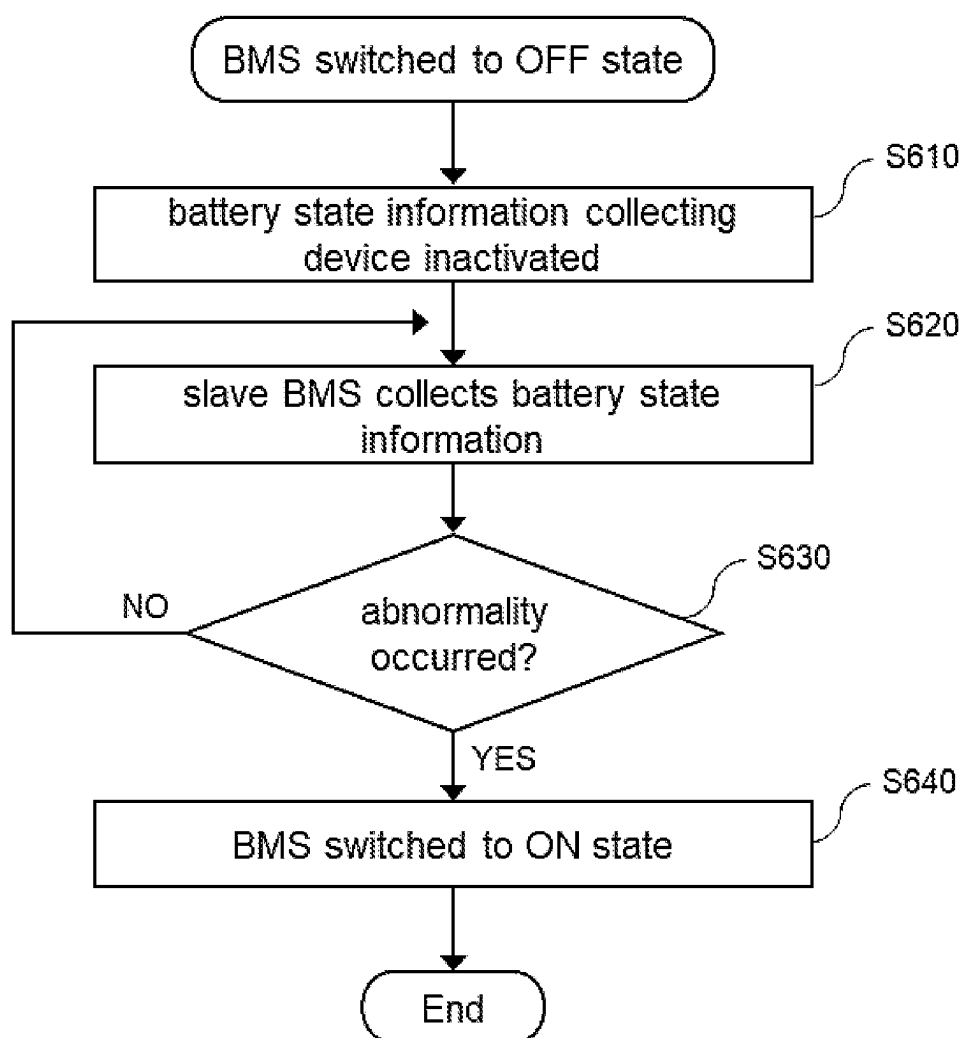
[Figure 4]



[Figure 5]



[Figure 6]



**BATTERY MANAGEMENT SYSTEM HAVING
HIERARCHICAL STRUCTURE AND
METHOD FOR OPERATING SAME****CROSS-REFERENCE TO RELATED
APPLICATIONS**

[0001] The present application is a national phase entry under 35 U.S.C. § 371 of International Application No. PCT/PCT/KR2023/008699 filed Jun. 22, 2023, which claims priority from Korean Patent Application No. 10-2022-0118506 filed in the Korean Intellectual Property Office on Sep. 20, 2022 and Korean Patent Application No. 10-2023-0074116 filed in the Korean Intellectual Property Office on Jun. 9, 2023, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present invention relates to a battery management system having a hierarchical structure and an operating method thereof, and more particularly, to a battery management system including a master BMS and a slave BMS and an operating method thereof.

BACKGROUND

[0003] A secondary battery is a battery that can be recharged and reused even after being discharged. The secondary battery can be used as an energy source for small devices such as mobile phones, tablet PCs and vacuum cleaners, and also used as an energy source for medium and large devices such as ESS (Energy Storage System) for automobiles and smart grids.

[0004] The secondary battery is applied to a system in a form of an assembly such as a battery module in which a plurality of battery cells are connected in series and parallel or a battery pack in which battery modules are connected in series and parallel according to system requirements. In a case of medium or large-sized devices such as electric vehicles, a high-capacity battery system in which a plurality of battery packs are connected in parallel may be applied in order to satisfy a required capacity of the device.

[0005] A battery management system that monitors and controls a state and operations of a battery pack may have a hierarchical structure in order to efficiently monitor and control all battery cells included in the battery pack. More specifically, the battery management system may include a plurality of slave BMSs that collect state information of battery cells or battery modules and a master BMS that collects state information of a battery pack, and receives cell/module state information from the slave BMSs for integrated monitoring and control.

[0006] In order for the hierarchical structured battery management system to perform state diagnosis more accurately, it is necessary to diagnose whether or not the battery is abnormal using several pieces of state information measured at the same time, but a device that collect pack state information and a device that collect information about cell or module state are configured separately and communication lines of respective devices are separated, so that a time error among several pieces of state information may occur. Due to this time error among the several pieces of state information, the reliability for diagnosis result of the battery management system may be limited.

SUMMARY**Technical Problem**

[0007] To obviate one or more problems of the related art, embodiments of the present disclosure provide a battery system having a hierarchical structure.

[0008] To obviate one or more problems of the related art, embodiments of the present disclosure also provide a method for operating a battery system having a hierarchical structure.

Technical Solution

[0009] In order to achieve the objective of the present disclosure, a battery management system (BMS) having a hierarchical structure may include a plurality of slave BMSs and a master BMS configured to interwork with the plurality of slave BMSs and to monitor a battery group of a plurality of batteries.

[0010] Here, the master BMS may be configured to collect state information of one or more batteries of the plurality of batteries from the plurality of slave BMSs and to collect state information of the battery group and the plurality of slave BMSs are configured to transmit the state information to the master BMS and are sequentially connected to the master BMS through a serial communication network.

[0011] The master BMS may include: a controller; and a communication interface. Here, the controller and the communication interface may be configured to be sequentially connected to the plurality of slave BMSs and the master BMS is configured to collect the group state information through the serial communication network.

[0012] The controller may be configured to receive the state information of the one or more batteries and the state information of the battery group through the communication interface, wherein the state information of the one or more batteries and the state information of the battery group are time-synchronized.

[0013] The controller may be configured to diagnose whether the one or more batteries or the battery group is abnormal based on the received state information of the one or more batteries and the state information of the battery group.

[0014] The communication interface may be directly connected to one slave BMS of the plurality of slave BMSs located at an end of the plurality of slave BMSs through a serial communication line.

[0015] In response to the battery management system being switched to an OFF state, the master BMS is configured to cease the collection of the group state information and the plurality of slave BMSs are configured to wake up at a predefined time.

[0016] The plurality of slave BMSs may be configured to wake up at intervals separated by the predefined time to collect the state information of the one or more batteries and diagnose whether each of the one or more batteries is abnormal based on the collected state information.

[0017] In response to the one or more batteries being diagnosed to be abnormal, one or more slave BMSs corresponding to the one or more batteries may be configured to transmit an abnormality signal to the communication interface through the serial communication line and the commu-

nication interface may be configured to transmit a wake-up signal to the controller to switch the battery management system to an ON state.

[0018] The state information of the battery group may include one or more information from among a voltage value, a current value, and an insulation resistance value of the battery group, and the state information of the one or more batteries may include one or more information from among voltage values, current values, and temperature values of battery cells or battery subgroups.

[0019] The master BMS may be configured to collect the group state information as an integration of information including a group voltage information, a group current information and an insulation resistance information.

[0020] According to another embodiment of the present disclosure, an operating method of a battery management system (BMS), which includes a plurality of slave BMSs and a master BMS sequentially connected with the plurality of slave BMSs through a serial communication network, may include: collecting, by the plurality of slave BMSs, state information of one or more batteries; collecting state information of a battery group of a plurality of batteries; receiving, by a controller of the master BMS, state information of the one or more batteries and state information of the battery group through a communication interface of the master BMS; and monitoring abnormality in the one or more batteries or in the battery group based on the collected state information.

[0021] The controller and the communication interface may be sequentially connected to the plurality of slave BMSs and the group state information is collected through the serial communication network.

[0022] The state information may include state information of the one or more batteries and state information of the battery group which are generated at a same time.

[0023] The monitoring abnormality may include diagnosing whether the one or more batteries or the battery group is abnormal based on the received state information of the one or more batteries and the state information of the battery group.

[0024] The communication interface may directly be connected to one slave BMS of the plurality of BMSs located at an end of the plurality of slave BMSs through a serial communication line.

[0025] The operating method of a battery management system may further include: in response to the battery management system being switched to an OFF state, collecting, by the plurality of slave BMSs, state information of the at least one battery of the one or more batteries by activating the plurality of slave BMSs at predetermined unit time intervals; and diagnosing, by the plurality of slave BMSs, whether or not the one or more batteries are abnormal based on the collected state information.

[0026] The operating method of a battery management system may further include: in response to diagnosing the one or more batteries are abnormal, transmitting, by one or more slave BMSs corresponding to the one or more batteries, one or more abnormality signals to the communication interface through the serial communication line; and transmitting, by the communication interface, a wake-up signal to the controller to switch the battery management system to an ON state.

[0027] The state information of the battery group may include one or more information from among a voltage

value, a current value, and an insulation resistance value of battery groups, and the state information on the one or more batteries may include one or more information from among voltage values, current values, and temperature values of battery cells or battery subgroups.

Advantageous Effects

[0028] According to the embodiment of the present invention as described above, the battery management system can diagnose a battery system based on time-synchronized state information, thereby improving reliability of monitoring and diagnosis results by the battery management system.

[0029] In addition, monitoring and diagnosis can be stably performed even in an OFF state of the battery management system according to embodiments of the present invention.

BRIEF DESCRIPTION OF THE DRAWINGS

[0030] FIG. 1 is a block diagram of a general battery management system.

[0031] FIG. 2 is a block diagram of a battery management system according to embodiments of the present invention.

[0032] FIG. 3 is a block diagram illustrating an implementation example of a battery management system according to the present invention.

[0033] FIG. 4 is an operation flowchart of a method for operating a battery management system according to embodiments of the present invention.

[0034] FIG. 5 is a block diagram of a battery management system to which a serial communication line according to embodiments of the present invention is applied.

[0035] FIG. 6 is an operation flowchart of a method of operating the battery management system shown in FIG. 5.

[0036] 100: Master BMS

[0037] 110: controller

[0038] 120: communication interface device

[0039] 130: group state information collecting device

[0040] 200: slave BMS

[0041] 300: serial communication network

[0042] 400: serial communication line

DETAILED DESCRIPTION

[0043] The present invention may be modified in various forms and have various embodiments, and specific embodiments thereof are shown by way of example in the drawings and will be described in detail below. It should be understood, however, that there is no intent to limit the present invention to the specific embodiments, but on the contrary, the present invention is to cover all modifications, equivalents, and alternatives falling within the spirit and technical scope of the present invention. Like reference numerals refer to like elements throughout the description of the figures.

[0044] It will be understood that, although the terms such as first, second, A, B, and the like may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the present invention. As used herein, the term “and/or” includes combinations of a plurality of associated listed items or any of the plurality of associated listed items.

[0045] It will be understood that when an element is referred to as being “coupled” or “connected” to another element, it can be directly coupled or connected to the other element or an intervening element may be present. In contrast, when an element is referred to as being “directly coupled” or “directly connected” to another element, there is no intervening element present.

[0046] The terms used herein is for the purpose of describing specific embodiments only and are not intended to limit the present invention. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises”, “comprising”, “includes”, “including” and/or “having”, when used herein, specify the presence of stated features, integers, steps, operations, constitutional elements, components and/or combinations thereof, but do not preclude the presence or addition of one or more other features, integers, steps, operations, constitutional elements, components, and/or combinations thereof.

[0047] Unless otherwise defined, all terms used herein, including technical and scientific terms, have the same meanings as commonly understood by one skilled in the art to which the present invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having meanings that are consistent with their meanings in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0048] FIG. 1 is a block diagram of a general battery management system.

[0049] Referring to FIG. 1, the battery management system may include a master BMS 10 and a plurality of slave BMSs 20.

[0050] The master BMS 10 is an upper layer BMS that monitors and controls a battery system including a plurality of battery modules #1 to #N. The master BMS 10 may include a plurality of pack state information collecting devices that collect state information of the battery pack. As shown in FIG. 1, the state information collecting device may include a pack voltage information collecting module, a pack current information collecting module, and an insulation resistance information collecting module.

[0051] The plurality of slave BMSs 20 each of which is a lower layer BMS are provided in correspondence with the plurality of battery modules #1 to #N, collect state information on the corresponding battery modules, and transmit the collected state information to the master BMS 10. Here, each of the slave BMSs may include a device for collecting module state information including one or more information of a voltage value, a current value, and a temperature value of each cell included in the battery module, and a voltage value, a current value, and a temperature value of each battery module, or each of the slave BMSs may be connected to the device for collecting module state information.

[0052] The master BMS 10 may include a communication interface device, and may be configured to transmit/receive data with a plurality of slave BMSs 20 through the communication interface device. Here, the master BMS 10 may receive module state information (cell state information and module state information) from each of the plurality of slave BMSs 20 through the communication interface device.

[0053] Referring to FIG. 1, the master BMS 10 may be configured to include a controller that receives pack state

information from pack state information collecting devices (pack voltage information collecting module, pack current information collecting module, and insulation resistance information collecting module) provided therein, and receives module state information from each of the plurality of slave BMSs 20 through the communication interface. Here, the controller of the master BMS 10 may monitor the collected pack state information and module state information and diagnose whether the battery system is abnormal.

[0054] Referring to FIG. 1, in the case of a general battery management system, pack state information collecting devices and module state information collection devices are separately configured, and communication is performed while a communication line for collecting pack state information and a communication line for collecting module state information are separated. Accordingly, several pieces of state information collected by the controller are data with a time error that is not measured at the same time. Due to this time error among several pieces of state information, the reliability for diagnosis result of the battery management system may be limited.

[0055] The present invention is presented to solve these technical problems. Hereinafter, the structure and operation method of a battery management system according to various embodiments of the present disclosure will be described in detail with reference to FIGS. 2 to 6.

[0056] FIG. 2 is a block diagram of a battery management system according to embodiments of the present invention.

[0057] Referring to FIG. 2, a battery management system according to embodiments of the present invention may include a master BMS 100 and a plurality of slave BMS 200.

[0058] The master BMS 100 is an upper layer BMS that monitors and controls a battery system including a plurality of batteries.

[0059] The master BMS 100 may include a controller 110, a communication interface device 120 and a group state information collecting device 130.

[0060] The controller 110 may monitor states of one or more of the batteries and battery groups included in the battery system and diagnose abnormalities. Here, the battery group may mean a battery module, a battery pack, a battery rack, or a battery bank.

[0061] The controller 110 may diagnose whether or not there is an abnormality in one or more of the batteries and battery groups based on the state information on the battery group transmitted from the group state information collecting device and the state information on one or more batteries transmitted from the plurality of slave BMSs 200.

[0062] The communication interface device 120 may include a communication module that receives state information from the group state information collecting device 130 and the plurality of slave BMSs 200 and transmits the state information to the controller 110.

[0063] The group state information collecting device 130 may collect state information in units of battery groups and transmit the collected group state information to the communication interface device 120. Here, the group state information collecting device 130 may be configured as an integrated module including a group voltage information collecting module, a group current information collecting module, and an insulation resistance information collecting module. Other words, unlike a general battery management system as shown in FIG. 1, the group state information collecting device 130 according to embodiments of the

present invention may be configured as one integrated module including a group voltage information collecting module, a group current information collecting module, and an insulation resistance information collecting module, and configured to simultaneously transmit the collected group state information (e.g., voltage value, current value, and insulation resistance value of each battery pack unit) to the communication interface device 120.

[0064] The plurality of slave BMSs 200 are lower layer BMSs that manage one or more batteries, respectively. Here, each of the slave BMSs may collect state information on a corresponding battery to be managed and transmit collected battery state information to the master BMS 100.

[0065] Each of the slave BMSs may be configured to include a state information collecting device for collecting state information of a battery to be managed, or to be connected to the state information collecting device. Here, the battery state information collected by a slave BMS may include state information of one or more of a voltage value, a current value, and a temperature value of a battery cell or battery subgroup unit (e.g., module unit).

[0066] The plurality of slave BMSs 200 may be sequentially connected to the group state information collecting device 130 of the master BMS 100 through a serial communication network. In addition, the controller 110 and the communication interface device 120 of the master BMS 100 may be sequentially connected to the group state information collecting device 130 and the plurality of slave BMSs 200 through a serial communication network. Referring to FIG. 2, a controller 110, a communication interface device 120, a group state information collecting device 130, and a plurality of slave BMSs 200 may be sequentially connected through a serial communication network 300.

[0067] Each of the components 110, 120, and 130 of the master BMS 100 and each of the plurality of slave BMSs 200 may include a communication module for forming the serial connection network 300 shown in FIG. 2. For example, the serial connection network 300 according to embodiments of the present invention may correspond to a daisy-chain communication network, and each of the components 110, 120, 130, and 200 in the master BMS 100 may include a data transeiving module to form a daisy-chain communication network.

[0068] From the slave BMS (#N) at the end to the slave BMS (#1) at the top, each

[0069] collected battery state information may be transmitted to the controller 110 through the serial communication network 300. Furthermore, the group state information collecting device 130 may transmit the collected group state information to the controller 110 through the serial communication network 300. Here, the group state information collecting device 130, as described above, may be configured as an integrated module including a group voltage information collecting module, a group current information collecting module, and an insulation resistance information collecting module, and thus, group state information including voltage values, current values, and insulation resistance values in group units may be transmitted to the controller 110.

[0070] In an embodiment, the battery state information and the group state information may correspond to state information generated at the same time. According to this embodiment, the controller 110 may receive pieces of time-synchronized state information (e.g., battery cell state infor-

mation, module state information, and pack state information), monitor the battery system using the received state information, and diagnose abnormalities in the battery system. Accordingly, reliability of monitoring and diagnosis results by the battery management system may be improved.

[0071] FIG. 3 is a block diagram illustrating an implementation example of a battery management system according to the present invention. Specifically, FIG. 3 is an implementation example of a battery management system in which a master BMS corresponds to a pack BMS and a slave BMS corresponds to a module BMS. Meanwhile, the battery management system shown in FIG. 3 is an example for clear description of the present invention, and the scope of the present invention is not limited thereto.

[0072] Referring to FIG. 3, a battery management system according to embodiments of the present invention may include a master BMS 100' and a plurality of slave BMSs 200'.

[0073] The master BMS 100' may correspond to a pack BMS that monitors and controls a battery pack including a plurality of battery modules.

[0074] The master BMS 100' may include a controller 110', a communication interface device 120', and a pack state information collecting device 130'.

[0075] The controller 110' may monitor the state of the battery pack and diagnose abnormalities in the battery pack. Here, the controller 110' may diagnose whether one or more of the battery cell, the battery module, and the battery pack are abnormal based on one or more of the pack state information transmitted from the pack state information collecting device and the module state information transmitted from the plurality of slave BMSs 200'.

[0076] The communication interface device 120' may be configured to include a communication module that receives state information from the plurality of slave BMSs 200' and a pack state information collecting device 130', and transmits the received state information to the controller 110'.

[0077] The pack state information collecting device 130' may collect pack state information in units of battery packs and transmit the collected pack state information to the communication interface device 120'. Here, the pack state information collecting device 130' may be configured as an integrated module including a pack voltage information collecting module, a pack current information collecting module, and an insulation resistance information collecting module.

[0078] The plurality of slave BMSs 200' may correspond to module BMSs provided in correspondence with the plurality of battery modules #1 to #N, respectively. Here, the slave BMS may collect state information on the corresponding battery module and transmit the collected module state information to the master BMS 100'.

[0079] Each of the slave BMSs may include a module state information collecting device for collecting module state information, or may be configured to be connected to the module state information collection device. Here, the module state information may include one or more state information of a voltage value, a current value, and a temperature value of each cell included in the battery module, and a voltage value, a current value, and a temperature value of each battery module.

[0080] The plurality of slave BMSs 200' may be sequentially connected to the pack state information collecting device 130' of the master BMS 100' through a serial com-

munication network. In addition, the controller 110' and the communication interface device 120' of the master BMS 100' may be sequentially connected to the pack state information collecting device 130' and the plurality of slave BMSs 200' through a serial communication network.

[0081] Each of the components 110', 120', and 130' of the master BMS 100' and each of the plurality of slave BMSs 200' may include a communication module to form the serial connection network 300' shown in FIG. 3.

[0082] From the slave BMS (#N) at the end to the slave BMS (#1) at the top, each collected module state information may be transmitted to the controller 110' through the serial communication network 300'. In addition, the pack state information collecting device 130' may transmit the collected pack state information to the controller 110' through the serial communication network 300'. Here, the module state information and the pack state information may correspond to state information generated at the same time.

[0083] FIG. 4 is an operation flowchart of a method for operating a battery management system according to embodiments of the present invention.

[0084] The plurality of slave BMSs may collect state information on one or more batteries and the group state information collecting device of the master BMS may collect state information on battery groups (S410). Here, the state information on the battery group may include information on one or more of a voltage value, a current value, and an insulation resistance value in units of battery group (e.g., in units of pack). In addition, the state information on one or more batteries may include one or more of a voltage value, a current value, and a temperature value in units of battery cells or battery subgroups (e.g., in units of module). Here, the collected battery state information and group state information may correspond to time-synchronized state information generated at the same time.

[0085] Thereafter, the controller of the master BMS may receive battery state information and group state information (S420). More specifically, the plurality of slave BMSs and the group state information collecting device may transmit battery state information and group state information to the communication interface device through a serial communication network, and the communication interface device may transfer the delivered battery state information and group state information to the controller.

[0086] The controller may monitor and diagnose a battery or a battery group based on the received battery state information and group state information (S430).

[0087] FIG. 5 is a block diagram of a battery management system to which a serial communication line according to embodiments of the present invention is applied.

[0088] The battery management system in FIG. 5 shows a structure further including a serial communication line 400 in the battery management system shown in FIG. 2, the serial communication line 400 directly connecting the slave BMS located at the end and the communication interface device 120.

[0089] Referring to FIG. 5, the slave BMS (#N) positioned at the end, among the plurality of slave BMSs 200, may be configured to be directly connected to the communication interface device 120 through the serial communication line 400. According to this embodiment, the battery management system may monitor the battery system even when it is in an OFF state, and may be switched to an ON state when abnormality occurs in the battery system.

[0090] When the battery management system is switched to an OFF state, the group state information collecting device 130 may be configured to be switched to an inactive state, and the plurality of slave BMSs 200 to wake-up at a predefined time.

[0091] More specifically, when the battery management system becomes turned off, the group state information collecting device 130 may be configured not to collect group state information, and the plurality of slave BMSs 200 may be configured to receive power through a regular power supply, wake-up every pre-defined time, and collect battery state information (e.g., voltage value, current value, temperature value of a cell, voltage value, current value, temperature value, of a module, etc.).

[0092] The plurality of slave BMSs 200 may diagnose whether a battery under management is abnormal based on the collected battery state information. Here, when an abnormality occurs in one or more batteries (e.g., when a voltage of a specific cell or module exceeds a predefined threshold), one or more corresponding slave BMSs may generate an abnormality signal. Here, the corresponding slave BMS may transfer the generated abnormality signal to the communication interface device 120 through the serial communication line 400. In other words, in an OFF state of the battery management system, the plurality of slave BMSs 200 may monitor respective batteries being managed, and, when an abnormality occurs, the slave BMS managing the abnormal battery may transmit an abnormality signal to the communication interface device 120 through the serial communication line 400 (a direction opposite to the communication direction in ON state shown in FIG. 2).

[0093] The communication interface device 120 may be configured to operate by receiving power through a constant power supply in an OFF state of the battery management system.

[0094] The communication interface device 120 may be configured to switch the battery management system into an ON state when an abnormality occurs in one or more batteries. For example, the plurality of slave BMSs 200 diagnose whether a battery under management is abnormal, and may transmit an abnormality signal indicating an abnormality to the communication interface device 120 through the serial communication line 400 when an abnormality occurs. Here, upon receiving the abnormality signal, the communication interface device 120 may generate a wake-up signal using a wake-up circuit and transmit the wake-up signal to the controller 110. The controller 110 may receive the wake-up signal and turn the battery management system to ON state.

[0095] In other words, in OFF state of the battery management system, the battery system may be monitored through the communication interface device 120, the slave BMS 200, and a separate serial communication line 400, whereas, when an abnormality occurs, the battery management system is switched to ON state and the controller 110 may diagnose abnormality in the entire battery system.

[0096] FIG. 6 is an operation flowchart of a method of operating the battery management system shown in FIG. 5.

[0097] Referring to FIGS. 5 and 6, when the battery management system is turned OFF, the group state information collecting device may be turned into an inactive state (S610).

[0098] Thereafter, the plurality of slave BMSs may be supplied with power through a constant power supply,

wake-up every predefined time, and collect battery state information (e.g., a voltage value, a current value, and a temperature value of a cell, and a voltage value, a current value, and a temperature value of a module, etc.) (S620).

[0099] Each of the slave BMSs may diagnose whether or not the battery under management is abnormal based on the battery state information (S630).

[0100] When an abnormality occurs in one or more batteries (YES in S630), the communication interface device may switch the battery management system to an ON state (S640). For example, when an abnormality occurs in a specific battery module, the communication interface device may receive an abnormality signal from a slave BMS corresponding to the module generating abnormality. Thereafter, the communication interface device may generate a wake-up signal through a wake-up circuit and transmit the wake-up signal to the controller, and the controller may turn the battery management system into an ON state.

[0101] The operations of the method according to the embodiments of the present invention may be implemented as a computer-readable program or code on a computer-readable recording medium. The computer-readable recording medium includes all types of recording devices in which data readable by a computer system is stored. In addition, the computer-readable recording medium may be distributed in a network-connected computer system to store and execute computer-readable programs or codes in a distributed manner.

[0102] Although some aspects of the invention have been described in the context of the apparatus, it may also represent a description according to a corresponding method, wherein a block or apparatus corresponds to a method step or feature of a method step. Similarly, aspects described in the context of a method may also represent a feature of a corresponding block or item or a corresponding apparatus. Some or all of the method steps may be performed by (or using) a hardware device, such as, for example, a microprocessor, a programmable computer, or an electronic circuit. In some embodiments, one or more of the most important method steps may be performed by such an apparatus.

[0103] In the forgoing, the present invention has been described with reference to the exemplary embodiment of the present invention, but those skilled in the art may appreciate that the present invention may be variously corrected and changed within the range without departing from the spirit and the area of the present invention described in the appending claims.

1. A battery management system (BMS) having a hierarchical structure comprising:

- a plurality of slave BMSs; and
- a master BMS configured to interwork with the plurality of slave BMSs and to monitor a battery group of a plurality of batteries,

wherein the master BMS is configured to collect state information of one or more batteries of the plurality of batteries from the plurality of slave BMSs and to collect state information of the battery group, and

wherein the plurality of slave BMSs are configured to transmit the state information to the master BMS are sequentially connected to the master BMS through a serial communication network.

2. The battery management system of claim 1, wherein the master BMS includes:

a controller; and
a communication interface,

wherein the controller and the communication interface are sequentially connected to the plurality of slave BMSs and the master BMS is configured to collect group state information ed through the serial communication network.

3. The battery management system of claim 2, wherein the controller is configured to receive the state information of the one or more batteries and the state information of the battery group through the communication interface, wherein the state information of the one or more batteries and the state information of the battery group are time-synchronized.

4. The battery management system of claim 3, wherein the controller is configured to diagnose whether the one or more batteries or the battery group is abnormal based on the received state information of the one or more batteries and the state information of the battery group.

5. The battery management system of claim 2, wherein the communication interface is directly connected to one slave BMS of the plurality of slave BMSs located at an end of the plurality of slave BMSs through a serial communication line.

6. The battery management system of claim 5, wherein, in response to the battery management system being switched to an OFF state, the master BMS is configured to cease the collection of the group state information and the plurality of slave BMSs are configured to wake-up at a predefined time.

7. The battery management system of claim 6, wherein, in the OFF state of the battery management system, the plurality of slave BMSs are configured to wake at intervals separated by the predefined time to collect the state information of the one or more batteries and diagnose whether each of the one or more batteries is abnormal based on the collected state information.

8. The battery management system of claim 7, wherein, in response to the one or more batteries being diagnosed to be abnormal,

one or more slave BMSs corresponding to the one or more batteries are configured to transmits an abnormality signal to the communication interface through the serial communication line, and

the communication interface is configured to transmit a wake-up signal to the controller to switch the battery management system to an ON state.

9. The battery management system of claim 1, wherein the state information of the battery group includes one or more information from among a voltage value, a current value, and an insulation resistance value of the battery groups, and wherein the state information of the one or more batteries includes one or more information from among voltage values, current values, and temperature values of battery cells or battery subgroups.

10. The battery management system of claim 1, wherein master BMS is configured to collect the group state information as an integration of information including a group voltage information, a group current information and an insulation resistance information.

11. An operating method of a battery management system (BMS) including a plurality of slave BMSs and a master BMS sequentially connected with the plurality of slave BMSs through a serial communication network, the method comprising:

collecting, by the plurality of slave BMSs, state information of one or more batteries;
collecting state information of a battery group of a plurality of batteries;
receiving, by a controller of the master BMS, state information of the one or more batteries and state information of the battery group through a communication interface of the master BMS; and
monitoring abnormality in the one or more batteries or in the battery group based on the collected state information.

12. The method of claim **11**, wherein the controller and the communication interface are sequentially connected to the plurality of slave BMSs and the group state information is collected through the serial communication network.

13. The method of claim **12**, wherein the state information includes the state information of the one or more batteries and the state information of the battery group, wherein the state information of the one or more batteries and the state information of the battery group are generated at a same time.

14. The method of claim **13**, wherein monitoring abnormality includes diagnosing whether the one or more batteries or the battery group is abnormal based on the received state information of the one or more batteries and the state information of the battery group.

15. The method of claim **12**, wherein the communication interface is directly connected to one slave BMS of the plurality of BMSs located at an end of the plurality of slave BMSs through a serial communication line.

16. The method of claim **15**, further comprising:
in response to the battery management system being switched to an OFF state,
collecting, by the plurality of slave BMSs, state information of the at least one battery of the one or more batteries by activating the plurality of slave BMSs at predetermined unit time intervals; and
diagnosing, by the plurality of slave BMSs, whether or not the one or more batteries are abnormal based on the collected state information.

17. The method of claim **16**, further comprising:
in response to diagnosing the one or more batteries are abnormal,
transmitting, by one ore more slave BMSs corresponding to the one or more batteries, one or more abnormality signals to the communication interface through the serial communication line; and
transmitting, by the communication interface, a wake-up signal to the controller to switch the battery management system to an ON state.

18. The method of claim **11**, wherein the state information of the battery group includes one or more information from among a voltage value, a current value, and an insulation resistance value of battery groups, and
wherein the state information on the one or more batteries includes one or more information from among voltage values, current values, and temperature values of battery cells or battery subgroups.

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